

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and deployed **Python 3.14 microservices** on **AWS Lambda** using **AWS CDK** for device control and cooking session management, handling approximately **1,500 events per second** during peak usage.
- Implemented event-driven patterns using **AWS EventBridge** and **SQS/SNS** to decouple backend services, ensuring reliable asynchronous communication for media and recipe data flows.
- Contributed to platform observability by integrating **structured logging** and **distributed tracing** with Datadog, improving incident root cause analysis time by **30%**.
- Participated in the full application lifecycle, including sprint planning, technical estimations, and code reviews for new platform capabilities, mentoring junior engineers on **Python best practices**.
- Executed cloud-to-cloud integrations with new appliance brand partners, establishing secure data exchange protocols and validating API contract adherence.
- Remained current with industry innovations, researching and presenting on new API gateway patterns and serverless deployment strategies to the engineering team.
- Managed production platform operations during an on-call rotation, responding to incidents and ensuring service health for a global user base.

SKILLS

Languages – Python, SQL, Java, R

Cloud Platforms – AWS (Lambda, EventBridge, SQS, SNS, CDK, EC2, S3, CloudWatch, IoT Core), GCP, Azure

Backend Frameworks & Libraries – Python 3.14, AWS Lambda Powertools, Pydantic v2, SQLAlchemy, Alembic, Valkey, OpenSearch, Elasticsearch, FastAPI, Flask, Java Spring Boot

Observability & Monitoring – Datadog, Splunk, CloudWatch, Dynatrace

Databases – PostgreSQL, MySQL, Oracle, DynamoDB, PynamoDB

Development Tools & Methodologies – Git, Jenkins, Docker, Kubernetes, AWS CDK, BDD, pytest, moto, Agile, Scrum, CI/CD, AI Code Generation

PROJECTS

IoT Device Control Service with Event-Driven Architecture

- Engineered a **Python microservice** on **AWS Lambda** to manage connected kitchen appliance states, processing over **1,000 state change events daily**.
- Integrated **AWS EventBridge** for asynchronous command dispatching and status updates, ensuring reliable communication between the platform and IoT devices.
- Implemented comprehensive **structured logging** and utilized **pytest** with **moto** for mocking AWS services, achieving **85% unit test coverage** for critical logic.

Smart Kitchen Platform Backend API Development

- Developed **RESTful APIs** using **Python (FastAPI)** to support mobile client interactions for recipe browsing and cooking session initiation, serving approximately **500 requests per minute**.
- Designed data models with **Pydantic v2** and integrated with **PostgreSQL** via **SQLModel** for persistent storage of user preferences and cooking data.
- Containerized the API service using **Docker** for consistent deployment across development and production environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)